Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2018 – Paper 2 – Solutions**

June 2018 – Question 1 – Solution

(a)(i) Let p and q be any two propositions. Complete the truth table below. [4]

$\boldsymbol{p}$	$\boldsymbol{q}$	$\sim\boldsymbol{p}$	$\sim\boldsymbol{q}$	$(\boldsymbol{p} \vee \boldsymbol{q})$	$\sim(\boldsymbol{p} \vee \boldsymbol{q})$	$\sim\boldsymbol{p} \wedge \sim\boldsymbol{q}$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

(ii) Hence, state whether the statements $\sim(\boldsymbol{p} \vee \boldsymbol{q})$ and $\sim\boldsymbol{p} \wedge \sim\boldsymbol{q}$ are logically equivalent. **Justify your response.** [1]

$\sim(p \vee q)$ and $\sim p \wedge \sim q$ are logically equivalent because their truth values are the same according to the last two columns in the truth table above.

(b) A binary operation $\oplus$ is defined on the set of real numbers, $\mathbb{R}$, as $a \oplus b = 2a + 3b$ for all a, b in $\mathbb{R}$.

(i) Calculate $5 \oplus 2$. [1]

$a \oplus b = 2a + 3b$

$5 \oplus 2 = 2(5) + 3(2)$

$= 10 + 6$

$= 16$

(ii) Prove that $\oplus$ is closed in $\mathbb{R}$. [3]

$\mathbb{R}$ is closed under $\oplus$ if $a \oplus b \in \mathbb{R}$, $\forall a, b \in \mathbb{R}$.

Since multiplication is closed under $\mathbb{R}$,

If $a \in \mathbb{R}$, then $2a \in \mathbb{R}$.

If $b \in \mathbb{R}$, then $3b \in \mathbb{R}$.

Since addition is closed under $\mathbb{R}$, then $2a + 3b \in \mathbb{R}$.

Hence, $\mathbb{R}$ is closed under $\oplus$.

Q.E.D.

(iii) Determine whether $\oplus$ is commutative. [3]

$a \oplus b = 2a + 3b$

$b \oplus a = 2b + 3a$

Since $2a + 3b \neq 2b + 3a$, then $a \oplus b \neq b \oplus a$.

Hence, $\oplus$ is not commutative.

(c) Given that $(2x+a)(x-1)(bx+1) \equiv cx^3 + 10x^2 - 2x - 10$ is a polynomial identity in x, calculate the values of a, b and c. [7]

Well,

$$(2x+a)(x-1)(bx+1) = (2x^2 - 2x + ax - a)(bx+1)$$

$$= 2bx^3 + 2x^2 - 2bx^2 - 2x + abx^2 + ax - abx - a$$

$$= 2bx^3 + (2 - 2b + ab)x^2 + (-2 + a - ab)x - a$$

Now,

$$2bx^3 + (2 - 2b + ab)x^2 + (-2 + a - ab)x - a = cx^3 + 10x^2 - 2x - 10$$

Comparing coefficients gives,

$2b = c$ → Equation 1

$2 - 2b + ab = 10$ → Equation 2

$-2 + a - ab = -2$ → Equation 3

$a = 10$ → Equation 4

Substituting $a = 10$ into Equation 2 gives:

$$2 - 2b + 10b = 10$$

$$2 + 8b = 10$$

$$8b = 8$$

$$b = \frac{8}{8}$$

$$b = 1$$

Substituting $b = 1$ into Equation 1 gives:

$2(1) = c$

$c = 2$

$\therefore a = 10, b = 1$ and $c = 2$.

(d) Solve the logarithmic equation $\log_4(2x+2) - \log_2(x+1) = 1$. [6]

$$\log_4(2x+2) - \log_2(x+1) = 1$$

$$\frac{\log_2(2x+2)}{\log_2 4} - \log_2(x+1) = 1$$

$$\frac{\log_2(2x+2)}{2} - \log_2(x+1) = 1$$

$$\log_2(2x+2) - 2\log_2(x+1) = 2$$

$$\log_2(2x+2) - \log_2(x+1)^2 = 2$$

$$\log_2 \frac{(2x+2)}{(x+1)^2} = 2$$

Converting to exponential form gives:

$$\frac{(2x+2)}{(x+1)^2} = 2^2$$

$$\frac{2(x+1)}{(x+1)(x+1)} = 4$$

$$\frac{2}{x+1} = 4$$

$$4(x+1) = 2$$

$$4x + 4 = 2$$

$$4x = 2 - 4$$

$$4x = -2$$

$$x = \frac{-2}{4}$$

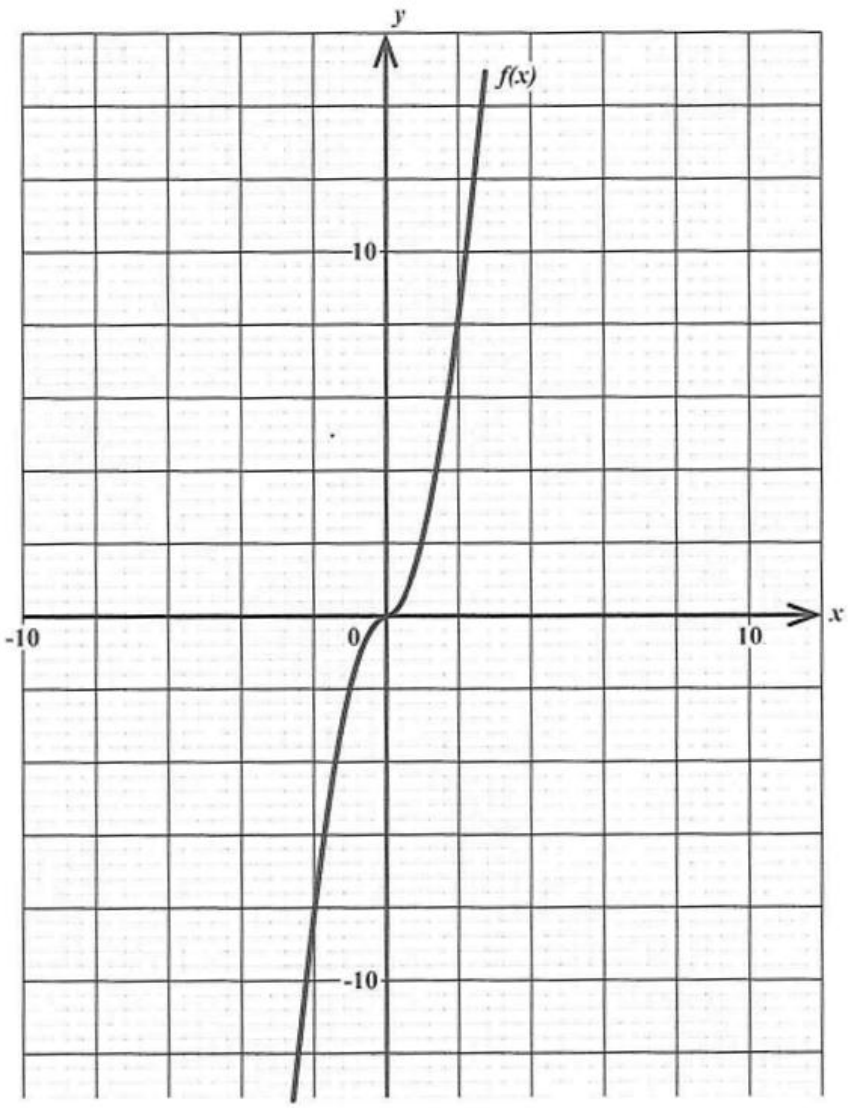
$$x = -\frac{1}{2}$$

$$\therefore x = -\frac{1}{2}$$

Total: 25 marks

June 2018 – Question 2 – Solution

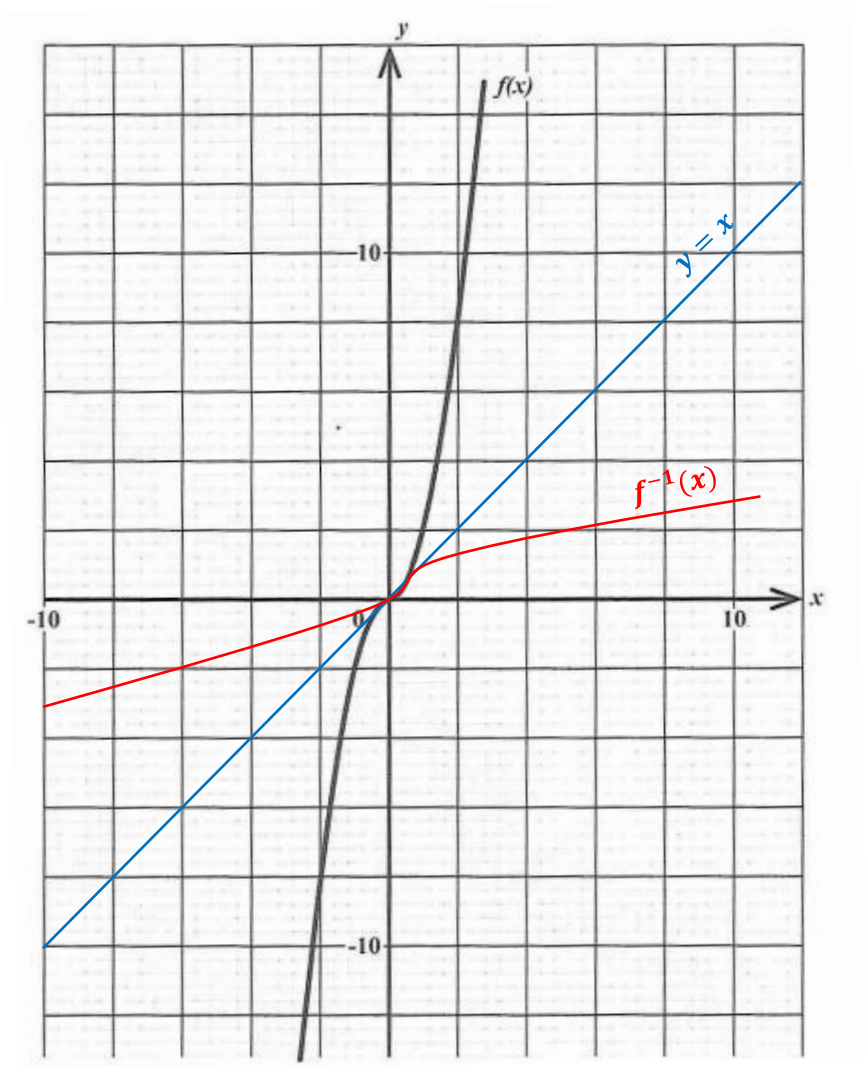
The diagram below shows the graph of the function $f(x)$.



(a) On the diagram above,

(i) sketch the inverse of $f(x)$ [2]

The graph of the inverse of $f^{-1}(x)$ is a reflection of the graph of $f(x)$ in the line $y = x$.



(ii) Use a graphical method to show that f is bijective. [3]

Any line parallel to the x-axis cuts the graph at most once.

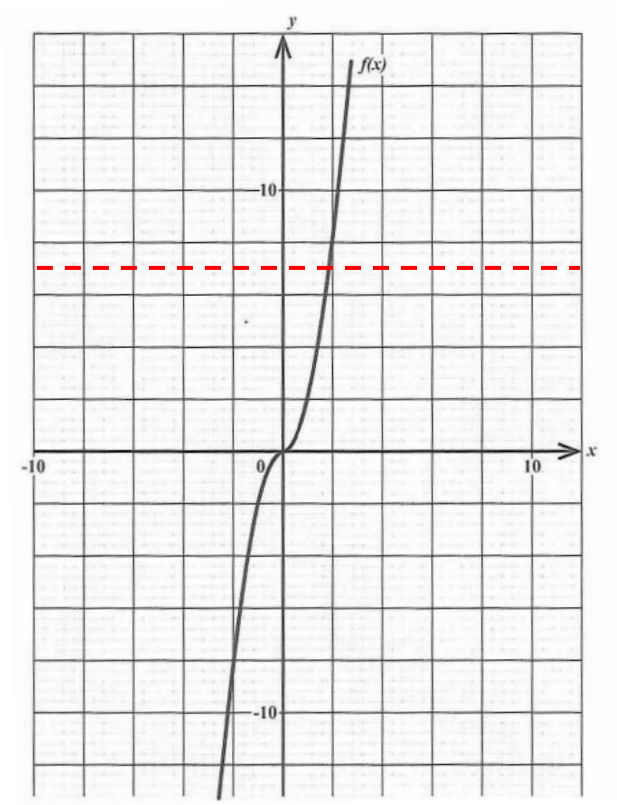
Hence, f is injective.

Any line parallel to the x-axis cuts the graph at least once.

Hence, f is surjective.

Since f is both injective and surjective, then f is bijective.

Q.E.D.



(b) (i) Prove that $|x-y| \leq |x-z| + |z-y|$ for all $x, y, z \in \mathbb{R}$. [4]

Proof:

$$x - y = x - z + z - y$$

$$(x-y) = (x-z) + (z-y)$$

$$\Rightarrow |x-y| = |(x-z) + (z-y)|$$

$$\Rightarrow |x-y| \leq \big||x-z| + |z-y|\big|$$

$$\Rightarrow |x-y| \leq |x-z| + |z-y|$$

Q.E.D.

(ii) Solve the inequality $|6x - 2| + x^2 \leq 5$. [8]

$|6x - 2| + x^2 \leq 5$

$|6x - 2| \leq 5 - x^2$

<u>Case 1:</u>

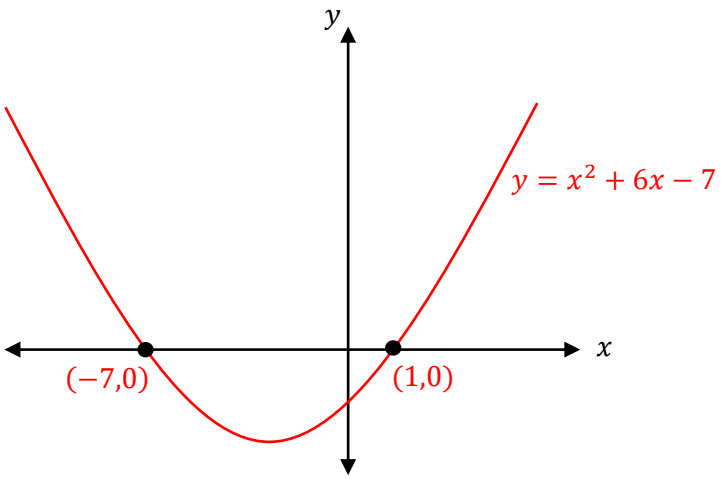
$6x - 2 \leq 5 - x^2$

$x^2 + 6x - 2 - 5 \leq 0$

$x^2 + 6x - 7 \leq 0$

$(x + 7)(x - 1) \leq 0$

The critical values are $x = -7$ and $x = 1$.



Since we are solving $x^2 + 6x - 7 \leq 0$, we look where the graph is negative.

The graph is negative for $-7 \leq x \leq 1$.

Case 2:

$$-(6x-2) \leq 5 - x^2$$

$$-6x + 2 \leq 5 - x^2$$

$$x^2 - 6x + 2 - 5 \leq 0$$

$$x^2 - 6x - 3 \leq 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

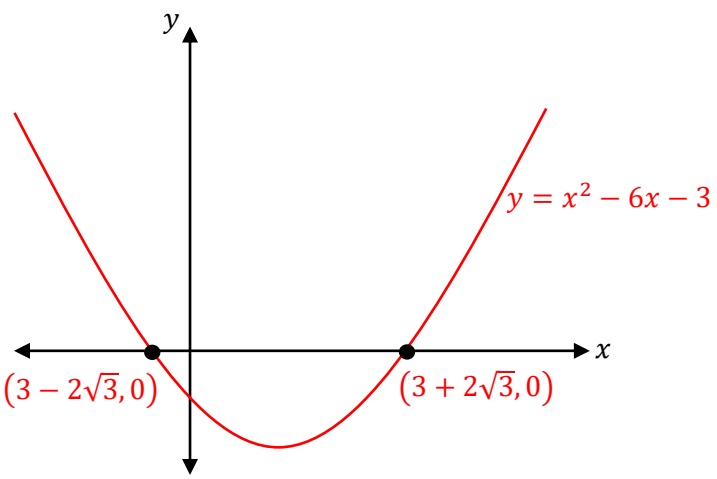
$$= \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(-3)}}{2(1)}$$

$$= \frac{6 \pm \sqrt{48}}{2}$$

$$= \frac{6 \pm 4\sqrt{3}}{2}$$

$$= 3 \pm 2\sqrt{3}$$

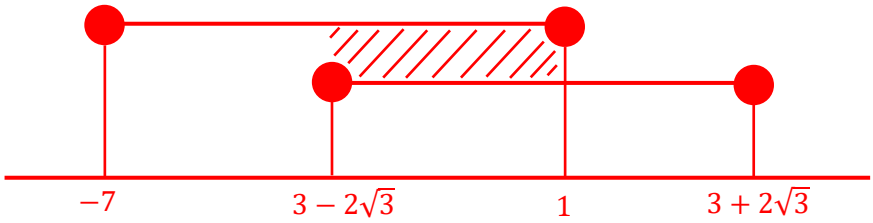
The critical values are $x = 3 - 2\sqrt{3}$ and $x = 3 + 2\sqrt{3}$



Since we are solving $x^2 - 6x - 3 \leq 0$, we look where the graph is negative.

The graph is negative for $3 - 2\sqrt{3} \leq x \leq 3 + 2\sqrt{3}$.

So, we have $-7 \leq x \leq 1$ and $3 - 2\sqrt{3} \leq x \leq 3 + 2\sqrt{3}$.



$\therefore$ The solution is $\{x: 3 - 2\sqrt{3} \leq x \leq 1\}$.

(c) Given that α, β and γ are the roots of the equation $2x^3 - x^2 + 1 = 0$, determine the equation whose roots are $\frac{1}{\alpha^2}$, $\frac{1}{\beta^2}$ and $\frac{1}{\gamma^2}$. [8]

Note: $(\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$= \frac{-(-1)}{2}$$

$$= \frac{1}{2}$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a}$$

$$= \frac{0}{2}$$

$$= 0$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$= -\frac{1}{2}$$

For the new roots:

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{(\beta\gamma)^2 + (\alpha\gamma)^2 + (\alpha\beta)^2}{\alpha^2\beta^2\gamma^2}$$

$$= \frac{(\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)}{(\alpha\beta\gamma)^2}$$

$$= \frac{0 - 2\left(-\frac{1}{2}\right)\left(\frac{1}{2}\right)}{\left(-\frac{1}{2}\right)^2}$$

$$= \frac{\frac{1}{2}}{\frac{1}{4}}$$

$$= \frac{1}{2} \div \frac{1}{4}$$

$$= \frac{1}{2} \times 4$$

$$= 2$$

$$\left(\frac{1}{\alpha^2}\right)\left(\frac{1}{\beta^2}\right) + \left(\frac{1}{\beta^2}\right)\left(\frac{1}{\gamma^2}\right) + \left(\frac{1}{\alpha^2}\right)\left(\frac{1}{\gamma^2}\right) = \frac{\gamma^2 + \alpha^2 + \beta^2}{\alpha^2\beta^2\gamma^2}$$

$$= \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)}{\alpha^2\beta^2\gamma^2}$$

$$= \frac{\left(\frac{1}{2}\right)^2 - 2(0)}{\left(-\frac{1}{2}\right)^2}$$

$$= 1$$

$$\left(\frac{1}{\alpha^2}\right)\left(\frac{1}{\beta^2}\right)\left(\frac{1}{\gamma^2}\right) = \frac{1}{(\alpha\beta\gamma)^2}$$

$$= \frac{1}{\left(-\frac{1}{2}\right)^2}$$

$$= \frac{1}{\frac{1}{4}}$$

$$= 1 \div \frac{1}{4}$$

$$= 1 \times 4$$

$$= 4$$

$\therefore$ The new equation is: $x^3 - 2x^2 + x - 4 = 0$.

Total: 25 marks

June 2018 – Question 3 – Solution

(a) (i) Show that $\frac{\sin 2\theta - \cos 2\theta + 1}{\cos 2\theta + \sin 2\theta - 1} = \sec 2\theta + \tan 2\theta$. [8]

Taking L.H.S:

$$\frac{\sin 2\theta - \cos 2\theta + 1}{\cos 2\theta + \sin 2\theta - 1} = \frac{\sin 2\theta - \cos 2\theta + 1}{\cos 2\theta + \sin 2\theta - 1} \times \frac{\cos 2\theta + \sin 2\theta + 1}{\cos 2\theta + \sin 2\theta + 1}$$

$$= \frac{(\sin 2\theta + 1 - \cos 2\theta)(\sin 2\theta + 1 + \cos 2\theta)}{(\cos 2\theta + \sin 2\theta - 1)(\cos 2\theta + \sin 2\theta + 1)}$$

$$= \frac{(\sin 2\theta + 1)^2 - \cos^2 2\theta}{(\cos 2\theta + \sin 2\theta)^2 - 1}$$

$$= \frac{\sin^2 2\theta + 2\sin 2\theta + 1 - \cos^2 2\theta}{\cos^2 2\theta + 2\cos 2\theta \sin 2\theta + \sin^2 2\theta - 1}$$

$$= \frac{\sin^2 2\theta + 2\sin 2\theta + (1 - \cos^2 2\theta)}{(\cos^2 2\theta + \sin^2 2\theta) + 2\cos 2\theta \sin 2\theta - 1}$$

$$= \frac{\sin^2 2\theta + 2\sin 2\theta + \sin^2 2\theta}{1 + 2\cos 2\theta \sin 2\theta - 1}$$

$$= \frac{2\sin^2 2\theta + 2\sin 2\theta}{2\cos 2\theta \sin 2\theta}$$

$$= \frac{2\sin 2\theta(\sin 2\theta + 1)}{2\sin 2\theta(\cos 2\theta)}$$

$$= \frac{\sin 2\theta + 1}{\cos 2\theta}$$

$$= \frac{\sin 2\theta}{\cos 2\theta} + \frac{1}{\cos 2\theta}$$

$$= \tan 2\theta + \sec 2\theta$$

$$= \sec 2\theta + \tan 2\theta$$

$$= \text{R.H.S.}$$

Alternative Method:

Taking L.H.S:

$$\frac{\sin 2\theta-\cos 2\theta+1}{\cos 2\theta+\sin 2\theta-1}=\frac{2\sin\theta\cos\theta-(1-2\sin^2\theta)+1}{(1-2\sin^2\theta)+2\sin\theta\cos\theta-1}$$

$$=\frac{2\sin\theta\cos\theta-1+2\sin^2\theta+1}{1-2\sin^2\theta+2\sin\theta\cos\theta-1}$$

$$=\frac{2\sin\theta\cos\theta+2\sin^2\theta}{2\sin\theta\cos\theta-2\sin^2\theta}$$

$$=\frac{2\sin\theta(\cos\theta+\sin\theta)}{2\sin\theta(\cos\theta-\sin\theta)}$$

$$=\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}$$

We multiply the numerator and denominator by the conjugate of the denominator.

We have,

$$\frac{\sin 2\theta-\cos 2\theta+1}{\cos 2\theta+\sin 2\theta-1}=\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\times\frac{\cos\theta+\sin\theta}{\cos\theta+\sin\theta}$$

$$=\frac{\cos^2\theta+2\cos\theta\sin\theta+\sin^2\theta}{\cos^2\theta-\sin^2\theta}$$

$$=\frac{1+\sin 2\theta}{\cos 2\theta}$$

$$=\frac{1}{\cos 2\theta}+\frac{\sin 2\theta}{\cos 2\theta}$$

$$=\sec 2\theta+\tan 2\theta$$

$$=\text{R.H.S.}$$

$$\therefore\ \frac{\sin 2\theta-\cos 2\theta+1}{\cos 2\theta+\sin 2\theta-1}=\sec 2\theta+\tan 2\theta$$

Q.E.D.

(ii) Hence, or otherwise, determine the general solution of $\frac{\sin\theta-\cos\theta+1}{\cos\theta+\sin\theta-1}=0$. [5]

$$\frac{\sin\theta-\cos\theta+1}{\cos\theta+\sin\theta-1}=0$$

$$\sec\theta+\tan\theta=0$$

$$\sec\theta=-\tan\theta$$

$$\frac{1}{\cos\theta}=-\frac{\sin\theta}{\cos\theta}$$

$$\cos\theta=-\sin\theta\cos\theta$$

$$\cos\theta\,(1+\sin\theta)=0$$

Either $\cos\theta=0$ or $\sin\theta+1=0$.

Consider $\cos\theta=0$.

$$PV=\frac{\pi}{2}$$

G.S. is $\theta=2n\pi\pm\frac{\pi}{2},\ \ n\in\mathbb{Z}$

Consider $\sin\theta+1=0$.

$$\sin\theta=-1$$

$$PV=-\frac{\pi}{2}$$

G.S. is $\theta = n\pi + (-1)^n\left(-\frac{\pi}{2}\right), \qquad n \in \mathbb{Z}$

$\theta = n\pi + (-1)^{n+1}\left(\frac{\pi}{2}\right), \qquad n \in \mathbb{Z}$

∴ The general solution is:

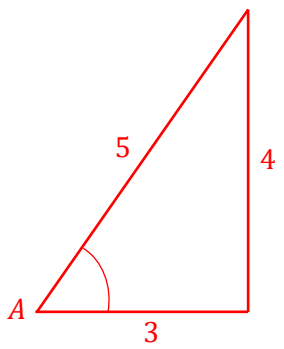
$\theta = 2n\pi \pm \frac{\pi}{2}, n \in \mathbb{Z}$ and $\theta = n\pi + (-1)^{n+1}\left(\frac{\pi}{2}\right), \qquad n \in \mathbb{Z}$

(b) Given that $\cos A = \frac{3}{5}$ and $\sin B = \frac{3}{4}$, where A and B are acute angles, calculate

(i) $\sin 2A$ [3]

We are given that $\cos A = \frac{3}{5}$

$= \frac{adj}{hyp}$



So,

$\sin A = \frac{opp}{hyp}$

$= \frac{4}{5}$

Now,

$\sin 2A = 2\sin A\cos A$

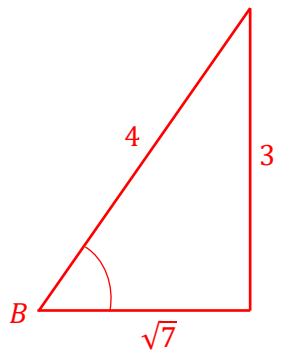
$= 2\left(\frac{4}{5}\right)\left(\frac{3}{5}\right)$

$= \frac{24}{25}$

(ii) $\cos(A+B)$ [3]

We are given that $\sin B = \frac{3}{4}$

$= \frac{opp}{hyp}$



$a^2 + b^2 = c^2$

$3^2 + b^2 = 4^2$

$9 + b^2 = 16$

$b^2 = 16 - 9$

$b^2 = 7$

$b = \sqrt{7}$

So,

$$\cos B = \frac{adj}{hyp}$$

$$= \frac{\sqrt{7}}{4}$$

Now,

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$= \left(\frac{3}{5}\right)\left(\frac{\sqrt{7}}{4}\right) - \left(\frac{4}{5}\right)\left(\frac{3}{4}\right)$$

$$= \frac{3\sqrt{7}}{20} - \frac{3}{5}$$

$$= \frac{3\sqrt{7}}{20} - \frac{12}{20}$$

$$= \frac{3\sqrt{7}-12}{20}$$

(c) Solve the equation $\sin\theta - \sqrt{3}\cos\theta = 1$, for $-\pi \leq \theta \leq \pi$. [6]

Let $\sin\theta - \sqrt{3}\cos\theta = r\sin(\theta - \alpha)$

$$= r\sin\theta\cos\alpha - r\cos\theta\sin\alpha$$

Equating coefficients gives:

$r\cos\alpha = 1$ $\rightarrow$ Equation 1

$r\sin\alpha = \sqrt{3}$ $\rightarrow$ Equation 2

Equation 2 $\div$ Equation 1 gives:

$$\frac{r\sin\alpha}{r\cos\alpha} = \frac{\sqrt{3}}{1}$$

$$\tan\alpha = \sqrt{3}$$

$$\alpha = \tan^{-1}(\sqrt{3})$$

$$\alpha = \frac{\pi}{3}$$

Now,

$$r = \sqrt{1^2 + (\sqrt{3})^2}$$

$$= \sqrt{1+3}$$

$$= \sqrt{4}$$

$$= 2$$

Hence, $\sin\theta - \sqrt{3}\cos\theta = 2\sin\left(\theta - \frac{\pi}{3}\right)$

We need to solve $2\sin\left(\theta - \frac{\pi}{3}\right) = 1$.

$$2\sin\left(\theta - \frac{\pi}{3}\right) = 1$$

$$\sin\left(\theta - \frac{\pi}{3}\right) = \frac{1}{2}$$

$$\theta - \frac{\pi}{3} = \frac{\pi}{6}, \pi - \frac{\pi}{6}$$

$+$	$\frac{\pi}{6}$

Note that:

$$-\pi \le \theta \le \pi$$

$$-\frac{4\pi}{3} \le \theta - \frac{\pi}{3} \le \frac{2\pi}{3}$$

$$\theta - \frac{\pi}{3} = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\theta - \frac{\pi}{3} = \frac{\pi}{6}, \frac{5\pi}{6} - 2\pi \qquad \left(\frac{5\pi}{6} \text{ is not in the interval, so check for other values}\right)$$

$$\theta - \frac{\pi}{3} = \frac{\pi}{6}, -\frac{7\pi}{6}$$

$$\theta = -\frac{7\pi}{6} + \frac{\pi}{3}, \frac{\pi}{6} + \frac{\pi}{3}$$

$$\theta = -\frac{5\pi}{6}, \frac{\pi}{2}$$

$\therefore \theta = -\frac{5\pi}{6}, \frac{\pi}{2}$ for $-\pi \leq \theta \leq \pi$.

Total: 25 marks

June 2018 – Question 4 – Solution

(a) The equations of the circle, C, and a straight line, L, are given by

$x^2 + y^2 - 2x + 2y + 1 = 0$ and $y = x - 3$, respectively.

(i) Determine the centre and radius of the circle, C. [3]

$$x^2 + y^2 - 2x + 2y + 1 = 0$$

$$x^2 - 2x + y^2 + 2y = -1$$

$$x^2 - 2x + 1 + y^2 + 2y + 1 = -1 + 1 + 1$$

$$(x - 1)^2 + (y + 1)^2 = 1$$

which is of the form $(x - a)^2 + (y - b)^2 = r^2$,

where the centre is $(1, -1)$ and radius, $r = 1$.

(ii) Show that $(1, -2)$ is one of the points of intersection of the circle, C, and the straight line, L. [6]

$x^2 + y^2 - 2x + 2y + 1 = 0$ → Equation 1

$y = x - 3$ → Equation 2

Substituting Equation 2 into Equation 1 gives:

$$x^2 + (x - 3)^2 - 2x + 2(x - 3) + 1 = 0$$

$$x^2 + x^2 - 6x + 9 - 2x + 2x - 6 + 1 = 0$$

$$2x^2 - 6x + 4 = 0$$

$$x^2 - 3x + 2 = 0$$

$$(x - 1)(x - 2) = 0$$

Either $x - 1 = 0$ or $x - 2 = 0$

$x = 1$ $x = 2$

When $x = 1$,

$y = 1 - 3$

$= -2$

$\therefore$ $(1, -2)$ is one of the points of intersection of the circle, C, and the straight line, L.

Q.E.D.

(iii) Determine the equation of the tangent to the circle, C, at the point $(1, -2)$. [4]

Consider points $(1, -2)$ and $(1, -1)$.

Gradient $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{-1-2}{1-1}$

$= \frac{-3}{0}$ (undefined)

The normal of the circle is a vertical line parallel to the y-axis.

Hence, the tangent is a horizontal line parallel to the x-axis.

$\therefore$ The equation of the tangent to the circle at the point $(1, -2)$ is: $y = -2$

(b) Let $\boldsymbol{u} = \begin{pmatrix} s \\ 3 \\ s \end{pmatrix}$ and $\boldsymbol{v} = \begin{pmatrix} -2 \\ 6 \\ 4 \end{pmatrix}$ be two position vectors in $\mathbb{R}^3$.

Show that $\boldsymbol{u}$ and $\boldsymbol{v}$ are NOT parallel. [9]

$\boldsymbol{u} = \begin{pmatrix} s \\ 3 \\ s \end{pmatrix}$, $\boldsymbol{v} = \begin{pmatrix} -2 \\ 6 \\ 4 \end{pmatrix}$

If $\boldsymbol{u}$ and $\boldsymbol{v}$ are parallel, $\boldsymbol{u} = t\boldsymbol{v}$.

So, we have,

$$\begin{pmatrix} s \\ 3 \\ s \end{pmatrix} = t\begin{pmatrix} -2 \\ 6 \\ 4 \end{pmatrix}$$

$$\begin{pmatrix} s \\ 3 \\ s \end{pmatrix} = \begin{pmatrix} -2t \\ 6t \\ 4t \end{pmatrix}$$

Equating corresponding entries gives:

$s = -2t$ $\rightarrow$ Equation 1

$3 = 6t$ $\rightarrow$ Equation 2

$s = 4t$ $\rightarrow$ Equation 3

From Equation 2:

$3 = 6t$

$t = \frac{3}{6}$

$t = \frac{1}{2}$

Substituting $t = \frac{1}{2}$ into Equation 1 gives:

$s = -2\left(\frac{1}{2}\right)$

$= -1$

Substituting $t = \frac{1}{2}$ into Equation 3 gives:

$s = 4\left(\frac{1}{2}\right)$

$= 2$ which is not possible

There does not exist a value for t such that $\boldsymbol{u} = t\boldsymbol{v}$.

Hence, $\boldsymbol{u}$ and $\boldsymbol{v}$ are not parallel.

Q.E.D.

(c) Determine the vector equation of a plane which passes through the point $(1, 3, 0)$ and which is perpendicular to the vector $2\boldsymbol{i} + 4\boldsymbol{j} + 5\boldsymbol{k}$. [3]

Let $\boldsymbol{a} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$, $\boldsymbol{n} = \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$

The vector equation of the plane is:

$$\boldsymbol{r} \cdot \boldsymbol{n} = \boldsymbol{a} \cdot \boldsymbol{n}$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix} = (1)(2) + (3)(4) + (0)(5)$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix} = 2 + 12 + 0$$

$$\boldsymbol{r} \cdot \begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix} = 14$$

Total: 25 marks

June 2018 – Question 5 – Solution

(a) Use the substitution $u = x^4 + 2$ to determine $\int (x^4 + 2)^3 (4x^7)\, dx$. [6]

Let $u = x^4 + 2$

$\frac{du}{dx} = 4x^3$

$dx = \frac{du}{4x^3}$

Also, $x^4 = u - 2$

Now,

$$\int (x^4 + 2)^3 (4x^7)\, dx = \int u^3 \cdot x^4 \cdot 4x^3 \, \frac{du}{4x^3}$$

$$= \int u^3 (u - 2)\ du$$

$$= \int (u^4 - 2u^3)\ du$$

$$= \frac{u^5}{5} - \frac{2u^4}{4} + c$$

$$= \frac{u^5}{5} - \frac{u^4}{2} + c$$

$$= \frac{(x^4+2)^5}{5} - \frac{(x^4+2)^4}{2} + c$$

(b) Calculate the area of the region lying between the parabolas $y = x^2$ and

$x = \frac{1}{8}y^2$. [6]

$y = x^2$ → Equation 1

$x = \frac{1}{8}y^2$ → Equation 2

Substituting Equation 1 into Equation 2 gives:

$$x = \frac{1}{8}(x^2)^2$$

$$x = \frac{1}{8}x^4$$

$$x - \frac{1}{8}x^4 = 0$$

$$x\left(1 - \frac{1}{8}x^3\right) = 0$$

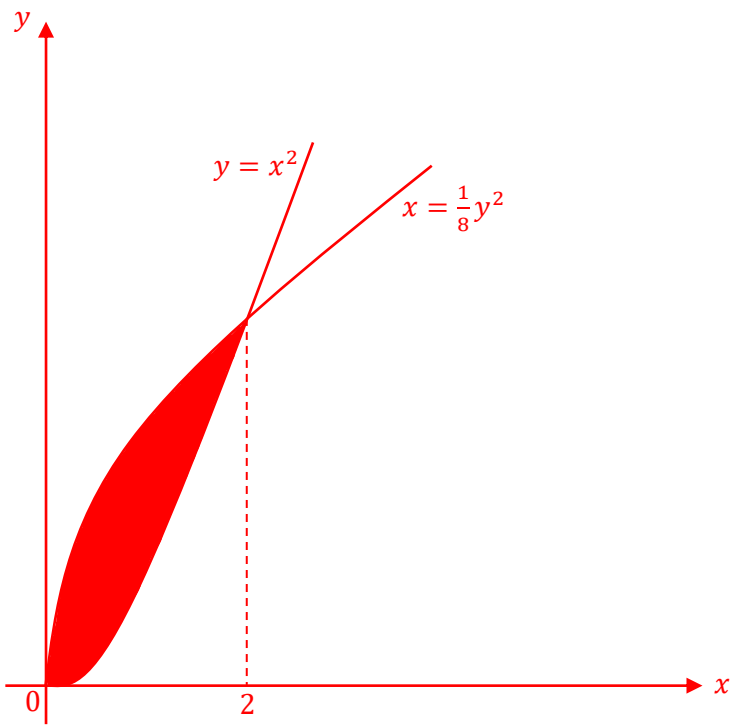
Either $x = 0$ or $1 - \frac{1}{8}x^3 = 0$

$$\frac{1}{8}x^3 = 1$$

$$x^3 = 8$$

$$x = 2$$

A sketch of the graphs is shown below:



Let Curve 1 be $y = x^2$ and let Curve 2 be $x = \frac{1}{8}y^2$.

Hence,

Area of shaded region = Area under Curve 2 − Area under Curve 1

Area under Curve 1 $= \int_a^b y\,dx$

$$= \int_0^2 x^2\,dx$$

$$= \left[\frac{x^3}{3}\right]_0^2$$

$$= \frac{1}{3}[(2)^3 - (0)^3]$$

$$= \frac{1}{3}(8 - 0)$$

$$= \frac{8}{3} \text{ units}^2$$

$x = \frac{1}{8}y^2$

$y^2 = 8x$

$y = \sqrt{8x}$

Area under Curve 2 $= \int_a^b \sqrt{8x}\, dx$

$= \sqrt{8} \int_0^2 x^{\frac{1}{2}}\, dx$

$= \sqrt{8} \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^2$

$= \frac{2\sqrt{8}}{3} \left[x^{\frac{3}{2}} \right]_0^2$

$= \frac{2\sqrt{8}}{3} \left[(2)^{\frac{3}{2}} - (0)^{\frac{3}{2}} \right]$

$= \frac{2\sqrt{8}}{3} \left[\sqrt{8} - 0 \right]$

$= \frac{2(8)}{3}$

$= \frac{16}{3}$ units^2

Therefore,

Area of shaded region = Area under Curve 2 − Area under Curve 1

$= \frac{16}{3} - \frac{8}{3}$

$= \frac{8}{3}$ units^2

(c) The equation of a curve is given as $f(x) = 3x^4 - 2x^3 - 6x^2 + 6x$.

(i) Determine f'. [1]

$f(x) = 3x^4 - 2x^3 - 6x^2 + 6x$

$f'(x) = 12x^3 - 6x^2 - 12x + 6$

(ii) Determine f''. [1]

$f'(x) = 12x^3 - 6x^2 - 12x + 6$

$f''(x) = 36x^2 - 12x - 12$

(iii) Calculate the x coordinates of the stationary points of

$$f(x) = 3x^4 - 2x^3 - 6x^2 + 6x$$

and determine the nature of these stationary points. [11]

$f'(x) = 12x^3 - 6x^2 - 12x + 6$

At stationary points, $f'(x) = 0$.

By trial and error, $f'(1) = 0$.

By the Factor Theorem, $(x - 1)$ is a factor of $f'(x)$.

When $f'(x) = 0$,

$$12x^3 - 6x^2 - 12x + 6 = 0$$

$$6(2x^3 - x^2 - 2x + 1) = 0$$

$$6(x-1)(ax^2 + bx + c) = 0$$

Now,

$$(x-1)(ax^2 + bx + c) = 2x^3 - x^2 - 2x + 1$$

$$ax^3 + bx^2 + cx - ax^2 - bx - c = 2x^3 - x^2 - 2x + 1$$

$$ax^3 + (b-a)x^2 + (c-b)x - c = 2x^3 - x^2 - 2x + 1$$

Comparing coefficients gives:

$a = 2$ → Equation 1

$b - a = -1$ → Equation 2

$c - b = -2$ → Equation 3

$-c = 1$ → Equation 4

Substituting $a = 2$ into Equation 2 gives:

$$b - 2 = -1$$

$$b = 2 - 1$$

$$b = 1$$

From Equation 3:

$-c = 1$

$c = -1$

We have $a = 2, b = 1$ and $c = -1$.

Hence,

$6(x-1)(ax^2 + bx + c) = 0$

$6(x-1)(2x^2 + x - 1) = 0$

$(x-1)(2x-1)(x+1) = 0$

Either $x - 1 = 0$ or $2x - 1 = 0$ or $x + 1 = 0$

$x = 1$ $x = \frac{1}{2}$ $x = -1$

Therefore, the x-coordinates of the stationary points are $x = -1, \frac{1}{2}$ and 1.

Now,

$f''(x) = 36x^2 - 12x - 12$

When $x = -1$,

$f''(-1) = 36(-1)^2 - 12(-1) - 12$

$= 36 + 12 - 12$

$= 36 \quad (> 0)$

$\therefore f(x)$ has a minimum at $x = -1$.

When $x = \frac{1}{2}$,

$f''\left(\frac{1}{2}\right) = 36\left(\frac{1}{2}\right)^2 - 12\left(\frac{1}{2}\right) - 12$

$= 9 - 6 - 12$

$= -9 \quad (< 0)$

$\therefore f(x)$ has a maximum at $x = \frac{1}{2}$.

When $x = 1$,

$f''(1) = 36(1)^2 - 12(1) - 12$

$= 36 - 12 - 12$

$= 12 \quad (> 0)$

$\therefore f(x)$ has a minimum at $x = 1$.

Total: 25 marks

June 2018 – Question 6 – Solution

(a) A function f is defined as

$$f(x) = \begin{cases} \frac{x^4-1}{x-1} & x < 1 \\ 4x & x > 1 \\ 2 & x = 1 \end{cases}$$

(i) Determine whether or not the limit of f at $x = 1$ exists. [4]

The limit of f exists at $x = 1$ iff $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x)$

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} \frac{x^4-1}{x-1}$$

$$= \lim_{x\to 1^-} \frac{(x-1)(x^3+x^2+x+1)}{(x-1)}$$

$$= \lim_{x\to 1^-} (x^3 + x^2 + x + 1)$$

$$= (1)^3 + (1)^2 + (1) + 1$$

$$= 4$$

Now,

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} 4x$$

$$= 4(1)$$

$$= 4$$

Since $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) = 4$, then the limit of f at $x = 1$ exists.

$\therefore \lim_{x\to 1} f(x) = 4$

(ii) Determine whether f is continuous at $x = 1$. [2]

Well, $f(1) = 2$ and $\lim_{x \to 1} f(x) = 4$.

Since $f(1) \neq \lim_{x \to 1} f(x)$, then the function is not continuous at $x = 1$.

(b) The parametric equation of a curve, g, is given by $x = 2\cos\theta$ and $y = 3 - \sin\theta$.

(i) Determine $\frac{d}{dx}\, g(x)$, in terms of θ. [3]

$x = 2\cos\theta$, $y = 3 - \sin\theta$

$\frac{dx}{d\theta} = -2\sin\theta$ $\frac{dy}{d\theta} = -\cos\theta$

Now,

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= -\cos\theta \times \frac{1}{-2\sin\theta}$$

$$= \frac{1}{2}\frac{\cos\theta}{\sin\theta}$$

$$= \frac{1}{2}\cot\theta$$

$$\therefore \frac{d}{dx}\, g(x) = \frac{1}{2}\cot\theta$$

(ii) Determine the equation of the normal to the curve at the point $\left(\sqrt{3}, \frac{5}{2}\right)$. [8]

$x = 2\cos\theta$

When $x = \sqrt{3}$,

$\sqrt{3} = 2\cos\theta$

$\cos\theta = \frac{\sqrt{3}}{2}$

$y = 3 - \sin\theta$

When $y = \frac{5}{2}$,

$\frac{5}{2} = 3 - \sin\theta$

$\sin\theta = 3 - \frac{5}{2}$

$\sin\theta = \frac{1}{2}$

$\frac{dy}{dx} = \frac{1}{2}\frac{\cos\theta}{\sin\theta}$

$= \frac{1}{2}\left(\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}\right)$

$= \frac{\sqrt{3}}{2}$

Gradient of tangent $= \frac{\sqrt{3}}{2}$

Gradient of normal $= -\frac{2}{\sqrt{3}}$

Substituting $m = -\frac{2}{\sqrt{3}}$ and point $\left(\sqrt{3}, \frac{5}{2}\right)$ into $y - y_1 = m(x - x_1)$ gives:

$y - \frac{5}{2} = -\frac{2}{\sqrt{3}}(x - \sqrt{3})$

$y - \frac{5}{2} = -\frac{2}{\sqrt{3}}x + 2$

$y = -\frac{2}{\sqrt{3}}x + 2 + \frac{5}{2}$

$y = -\frac{2}{\sqrt{3}}x + \frac{9}{2}$

$\therefore$ The equation of the normal to the curve is: $y = -\frac{2}{\sqrt{3}}x + \frac{9}{2}$

(c) The slope of a tangent of the curve, g, at any point (x, y) is equal to the product of the x-coordinate and the reciprocal of the y-coordinate of the point of tangency.

(i) Formulate an appropriate differential equation and find the equation of the curve family. [5]

$$\frac{dy}{dx} = x \times \frac{1}{y}$$

$$\frac{dy}{dx} = \frac{x}{y}$$

$$y\,dy = x\,dx$$

$$\int y\,dy = \int x\,dx$$

$$\frac{y^2}{2} = \frac{x^2}{2} + c$$

(ii) Hence, or otherwise, determine the equation of the curve given that it passes through the point $(1, 3)$. [3]

The equation of the curve passes through the point $(1, 3)$.

So, we have

$$\frac{(3)^2}{2} = \frac{(1)^2}{2} + c$$

$$\frac{9}{2} = \frac{1}{2} + c$$

$$x = \frac{9}{2} - \frac{1}{2}$$

$$x = 4$$

$\therefore$ The equation of the curve is: $\frac{y^2}{2} = \frac{x^2}{2} + 4$ or $y^2 = x^2 + 8$

Total: 25 marks